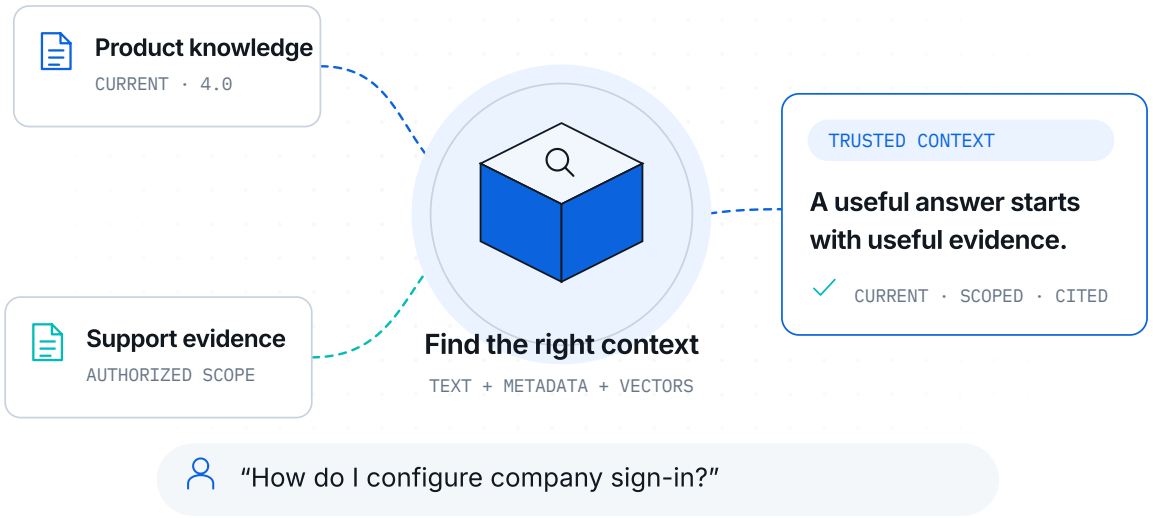
Elastic Search.
Trusted context.

Your users ask for an answer. Your platform must find the right context. Connect the knowledge, find what matters, and deliver with confidence.



CONNECT

The knowledge

Make current, authorized content retrievable.

FIND

What matters

Combine exact terms, intent, and measured relevance.

DELIVER

With confidence

Show evidence and evaluate the complete experience.

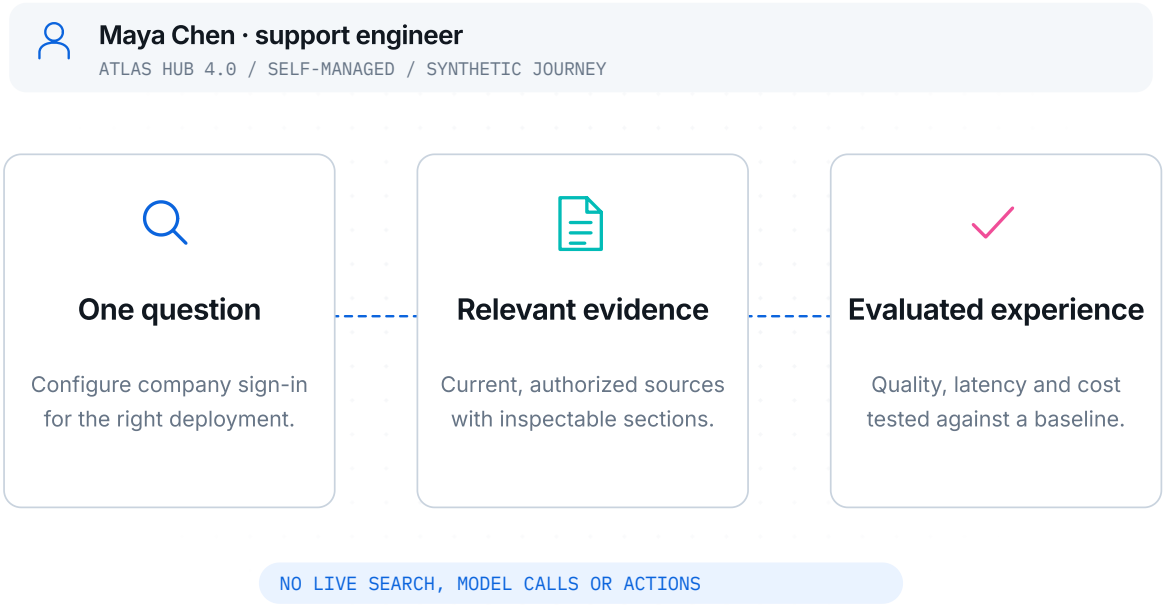
Original illustrative journey. No live queries, model calls, or actions. Feature requirements vary by version, subscription, model, and deployment.

SOURCES Search ↗

elastic

One question.
Traceable evidence.

How do I configure company sign-in for Atlas Hub 4.0 on a self-managed deployment?



SCOPE

The right documents

Atlas Hub 4.0, self-managed, current, and authorized.

RELEVANCE

The right passages

Keep exact identifiers while matching company sign-in to single sign-on.

EVIDENCE

An inspectable answer

Tie each prerequisite to a current source section.

Synthetic teaching example; no product benchmark or real configuration guidance. Eligibility is enforced before ranking and generation.

SOURCES RAG ↗ Document access ↗

elastic

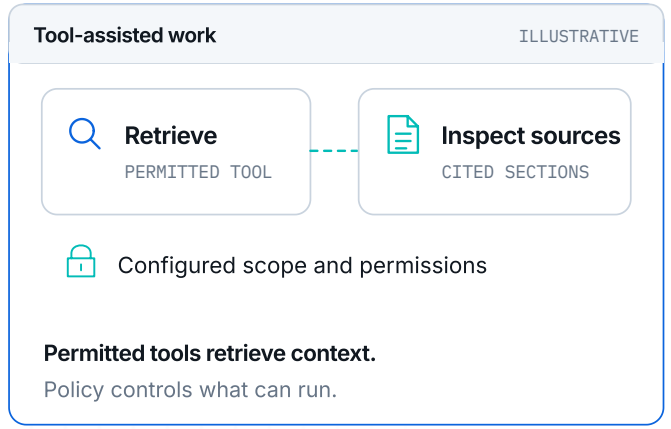
From finding links to finishing work.

Results, answers, and agents all depend on retrieval. A richer interface makes relevance and evidence more consequential.

USER EXPERIENCE EVOLVES

Relevance stays foundational.

1. Links
2. Relevant results
3. Grounded answers
4. **Tool-assisted work**



Tool-assisted work 4/4

RESULTS

A path to knowledge

Users inspect ranked documents and metadata.

ANSWERS

Context in the response

The application supplies evidence and source links.

AGENTS

Permitted assistance

Tools need configured privileges and evaluation.

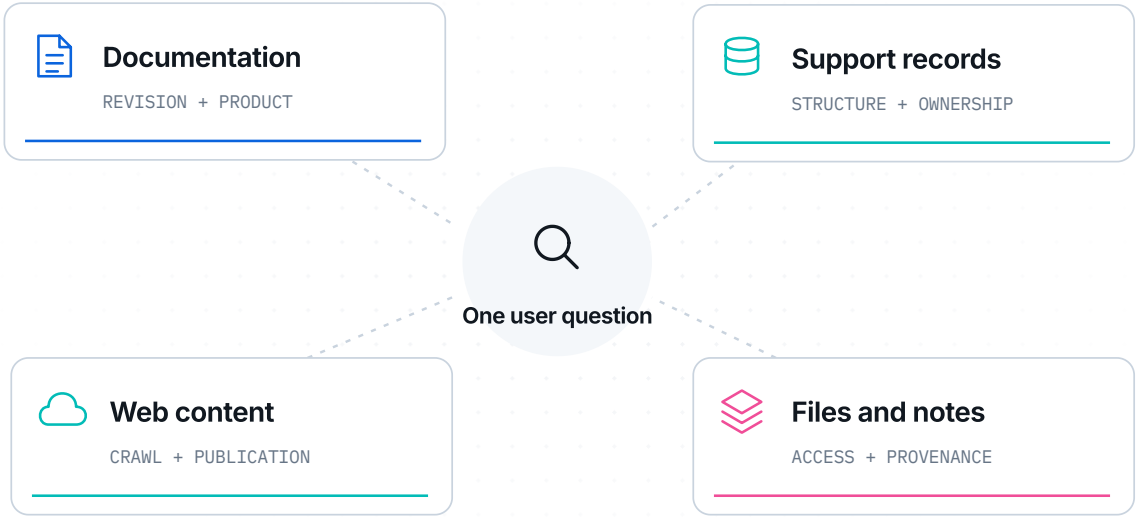
Illustrative progression. Grounded generation does not guarantee correctness, and tool availability does not confer action authority.

SOURCES Search ↗ RAG ↗ Agent Builder ↗

elastic

Knowledge has many homes. Users have one question.

Documents, records, web content, and access policies arrive from different systems. The user needs a coherent result, not that complexity.



CONTENT

Different formats

Extract searchable text and preserve useful structure.

CHANGE

Different update cycles

Sync revisions and retire superseded content.

ACCESS

Different boundaries

Map source ACLs and enforce retrieval permissions.

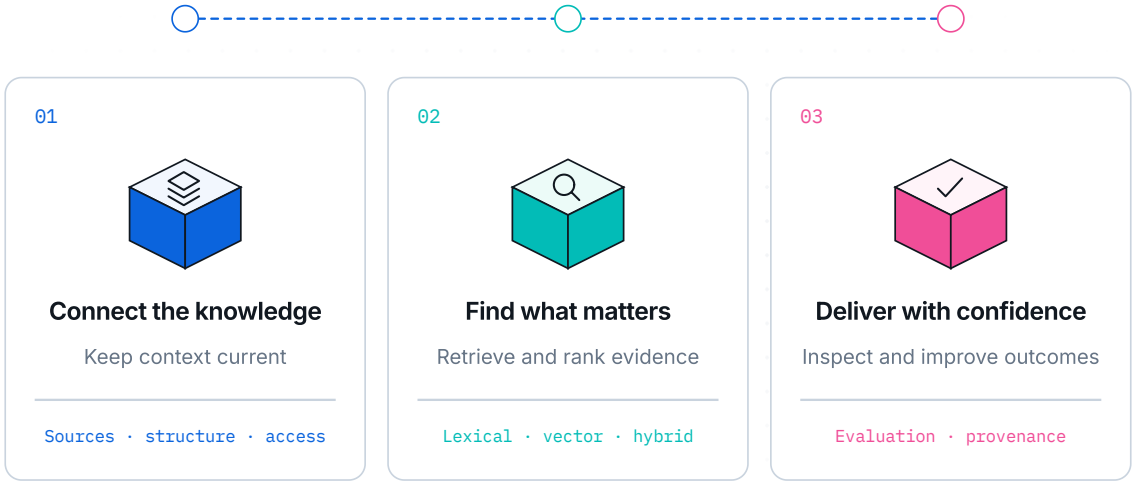
Connector coverage and ACL synchronization vary. Source permissions are not automatically inherited by every integration.

SOURCES Ingestion ↗ Document access ↗

elastic

Connect. Find. Deliver.

Connect the knowledge. Find what matters. Deliver with confidence. Each pillar answers a different failure mode.



ONE QUESTION. THREE ENGINEERING RESPONSIBILITIES.

CONNECT

Current and authorized

Engineer ingestion, metadata, revisions, and access.

FIND

Relevant and measurable

Compose retrieval and test representative questions.

DELIVER

Useful and verifiable

Expose evidence and operate within explicit limits.

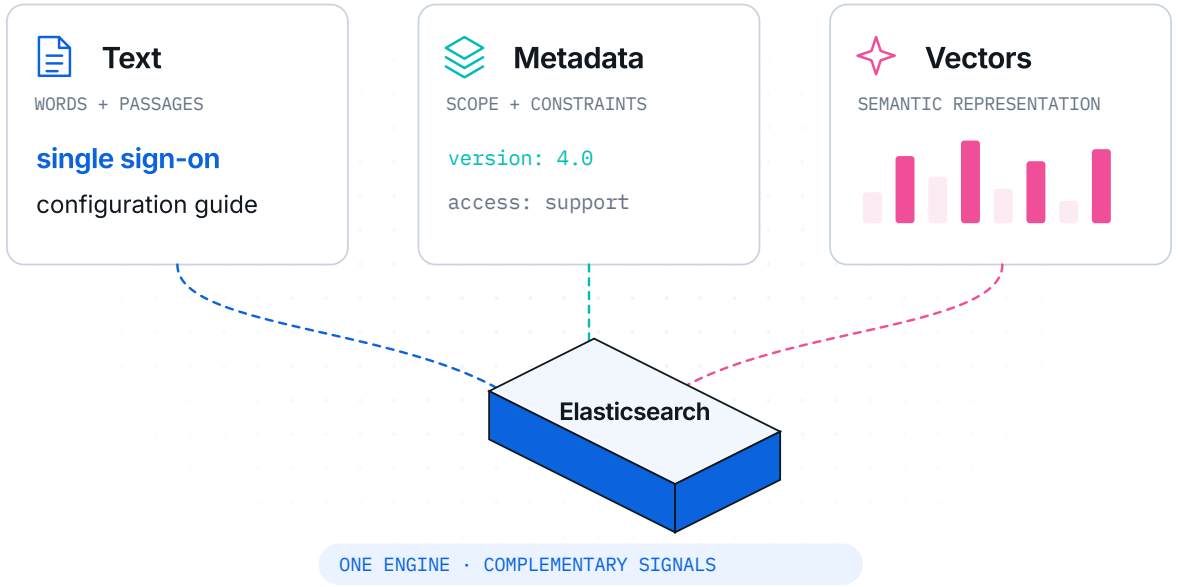
Architecture guidance. Outcomes depend on corpus quality, relevance design, application policy, and measured operations.

SOURCES Search ↗ RAG ↗

elastic

One engine. Several representations.

Text, structured metadata, and vectors work together. Exact identifiers, eligibility filters, and semantic similarity belong in the same retrieval design.



TEXT

Words and identifiers

Use full-text retrieval for terminology and exact intent.

METADATA

Explicit constraints

Filter version, deployment, freshness, and access scope.

VECTORS

Similarity signals

Add learned representations when they help the task.

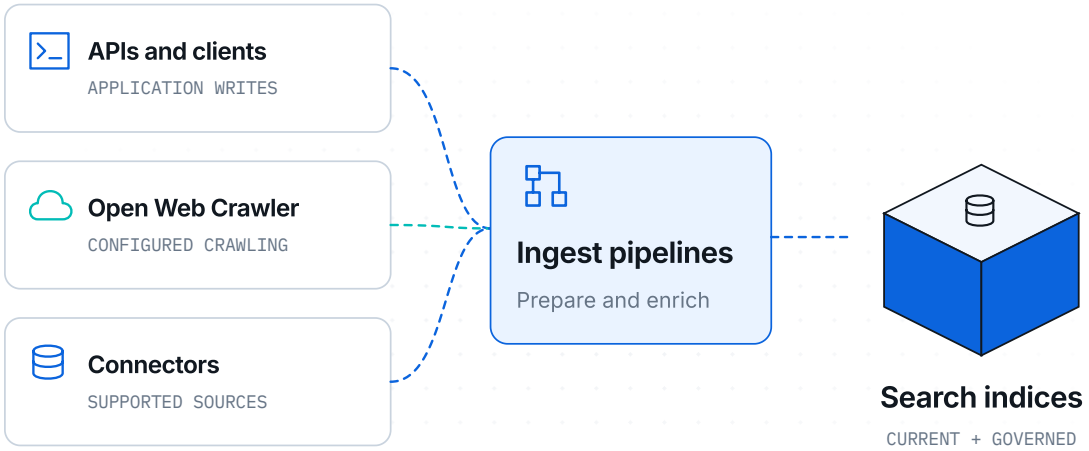
Near real-time search. Advanced retrieval and inference requirements vary by version, subscription, model, and deployment.

SOURCES Elasticsearch ↗ Retrievers ↗

elastic

Connect the sources. Build the pipeline.

Use APIs and clients, configured connectors, or the Open Web Crawler. Transform content and metadata before the first search request.



Schedule sync, manage failures, preserve revision and access metadata.

APIS

Application-owned data

Send JSON with official clients or Elasticsearch APIs.

CONNECTORS

Configured synchronization

Check source coverage, scheduling, and access support.

CRAWLER

Web content

Operate the Open Web Crawler for eligible pages.

Integration support and ACL sync vary. Cross-cluster search targets Elasticsearch indices with explicit remote access and connectivity.

SOURCES Ingestion ↗ Cross-cluster search ↗

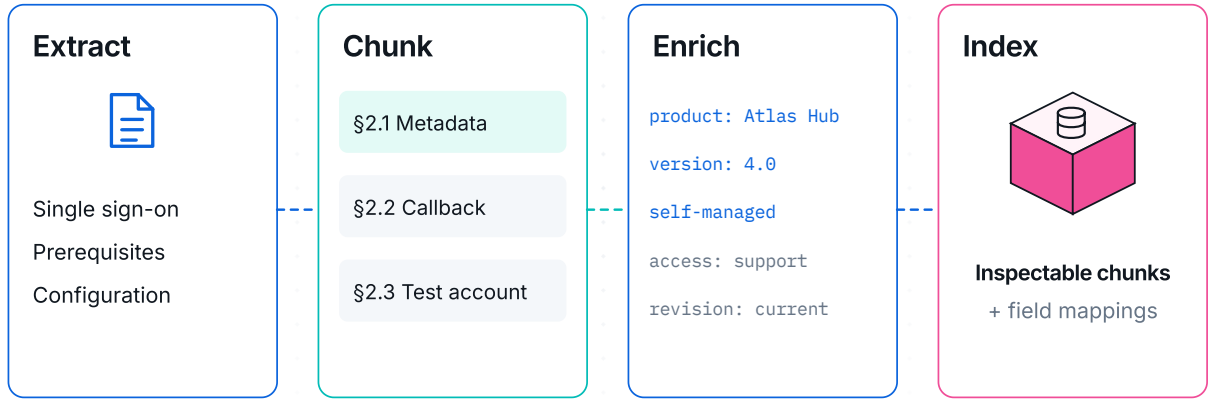
elastic

A file is not context yet.

Extract text, chunk useful passages, enrich metadata, and preserve source sections. Context should remain inspectable after indexing.

ATLAS HUB 4.0 · ADMINISTRATOR GUIDE

ILLUSTRATIVE DOCUMENT PREPARATION



Content, citation boundaries and permissions travel together.

Index 4/4

STRUCTURE

Useful passage boundaries

Keep context and source sections with each chunk.

METADATA

Scope before ranking

Record current version, deployment, and authorization.

INFERENCE

Configured representations

Pin the endpoint and evaluate model changes.

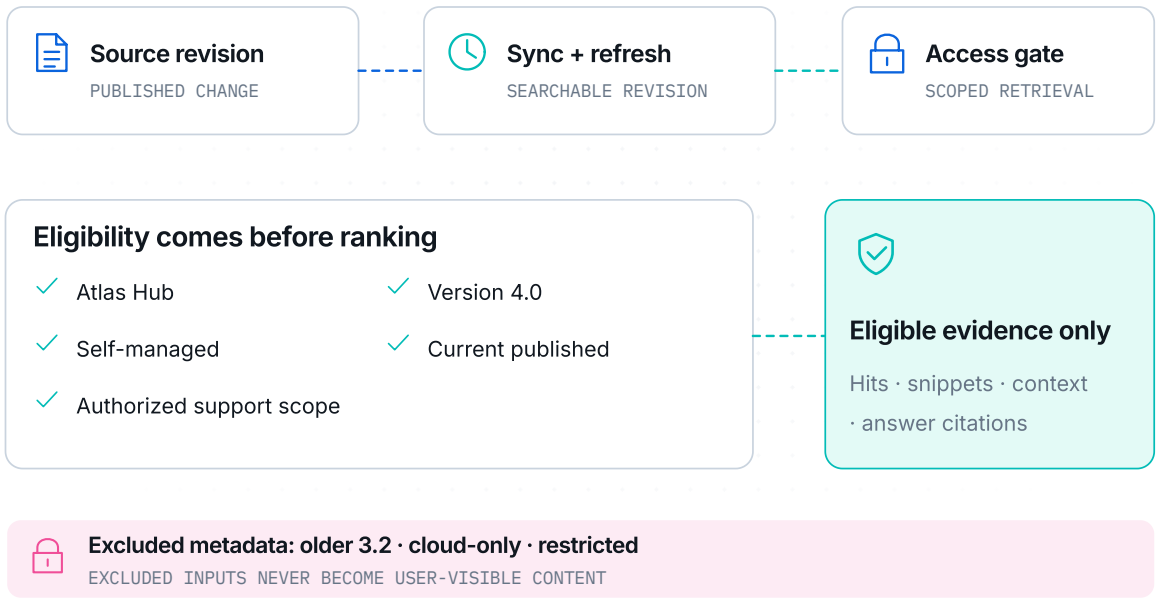
semantic_text accepts text, not arbitrary image or PDF bytes. Inference, license, model, language, and resource requirements apply.

SOURCES Ingestion ↗ semantic_text ↗

elastic

The right context. The right person and time.

Source changes, synchronization, refresh, and retrieval permissions form one path. Both lexical and vector branches use the same eligibility policy.



FRESHNESS

An engineered pipeline

Source sync and index refresh determine visibility.

ACCESS

Before model context

Enforce configured document and field permissions.

SCOPE

One shared policy

Exclude older, cloud-only, and restricted inputs.

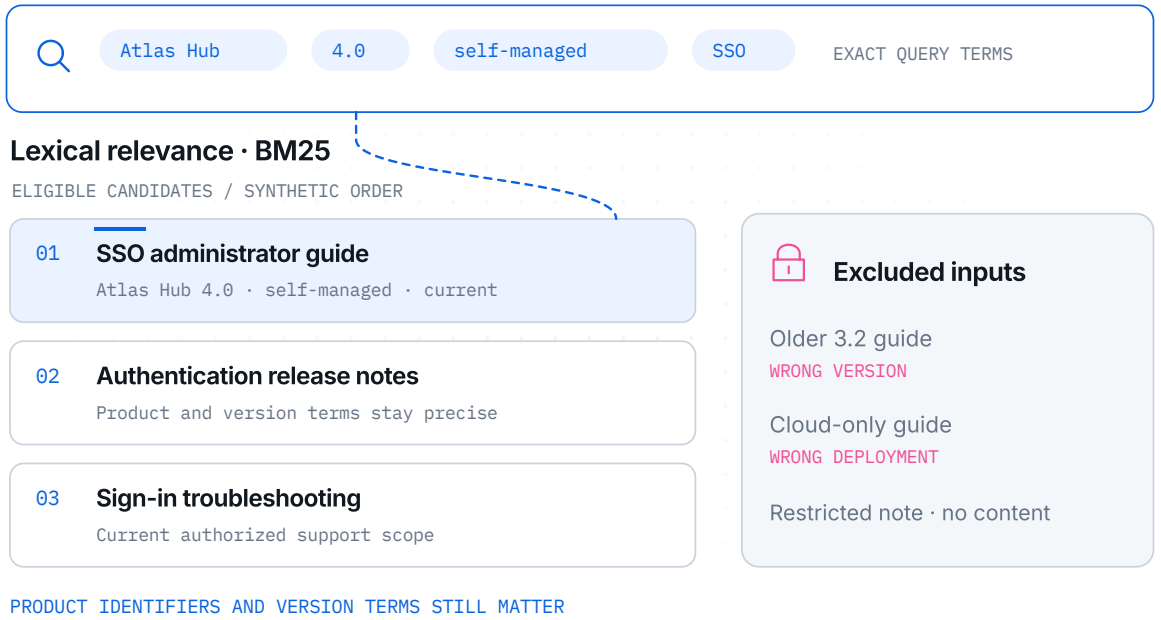
Configure ACL synchronization and Elasticsearch authorization. Prompts do not enforce access. DLS/FLS entitlements and limitations apply.

SOURCES Refresh ↗ Document access ↗

elastic

Exact words still matter.

Atlas Hub 4.0 self-managed SSO



IDENTIFIERS

Preserve the specifics

Atlas Hub and 4.0 are meaningful query terms.

SCORING

Lexical evidence

BM25 ranks text using statistical term signals.

ELIGIBILITY

Before the result

Apply the same current, deployment, and access scope.

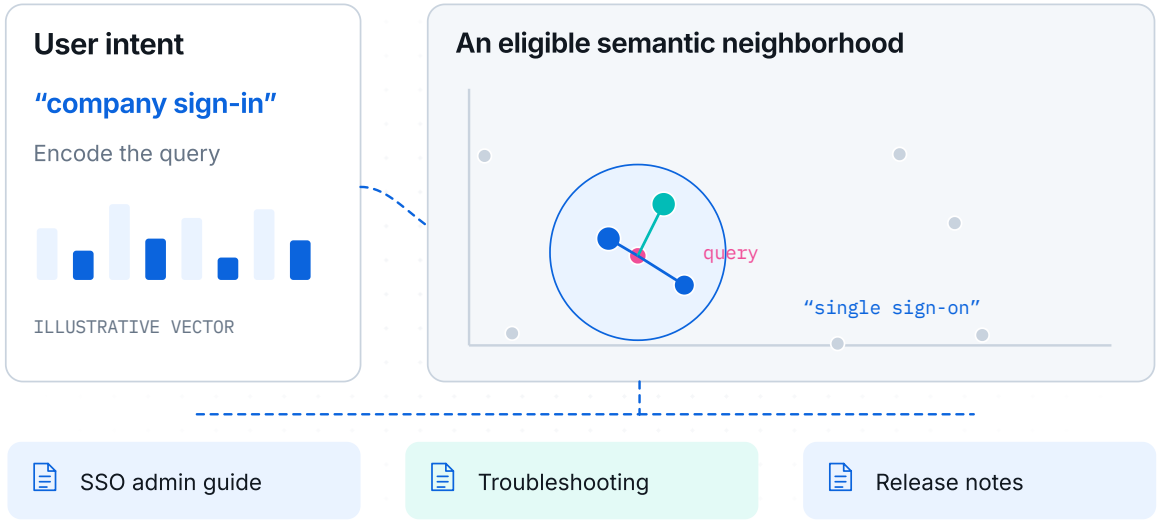
Rank positions are synthetic. Excluded inputs remain outside the result list. Retrieval quality depends on mappings, analyzers, corpus, and queries.

SOURCES Ranking ↗ Document access ↗

elastic

Intent has more than one wording.

Company sign-in and single sign-on can point to related context. Learned representations add a similarity signal within the same authorized scope.



ABSTRACT PROJECTION · SYNTHETIC POSITIONS · NOT A MODEL DIAGRAM

Return eligible context 3/3

INTENT

Related language

Retrieve beyond an exact word match when the model helps.

SCOPE

The same constraints

Similarity never overrides version, deployment, or access.

CHOICE

Evaluate the model

Test language, domain, latency, and representation changes.

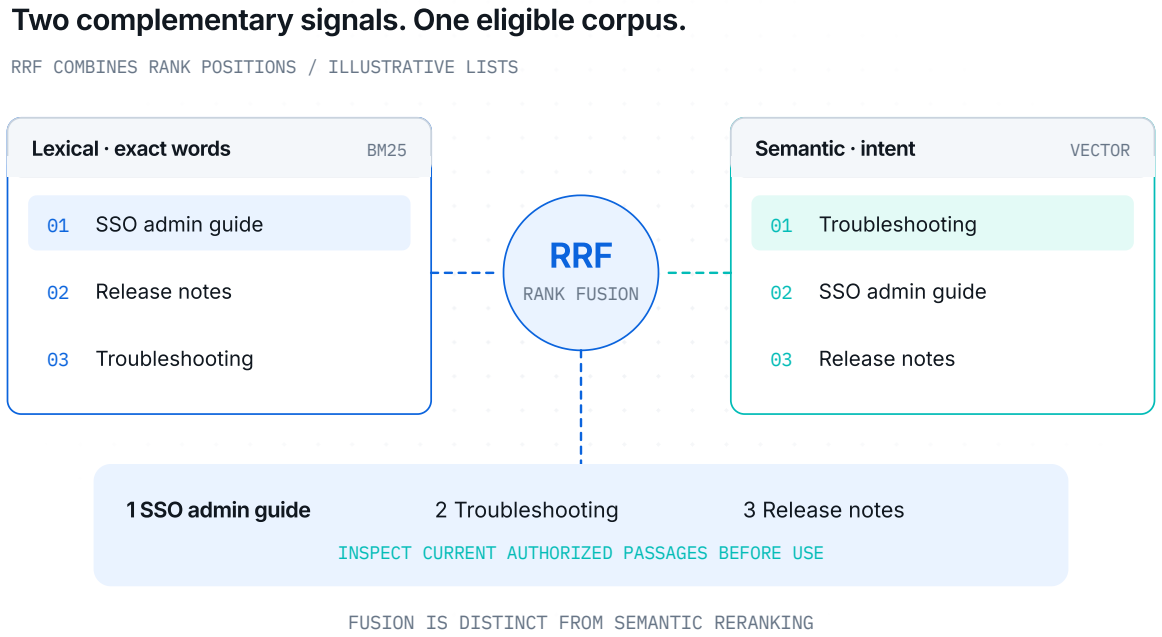
Similarity is a model-dependent signal. Embedding, language, hardware, API, version, and subscription requirements vary.

SOURCES Ranking ↗ semantic_text ↗

elastic

Two perspectives. One ranked result.

Lexical retrieval preserves terminology. Vector retrieval adds similarity. Reciprocal rank fusion combines their eligible ranked lists.



LEXICAL

Exact terms

Keep identifiers and domain wording in the design.

VECTOR

Similarity

Add candidates that match the user's phrasing.

RRF

Fuse ranked lists

Use rank positions rather than comparing raw scores.

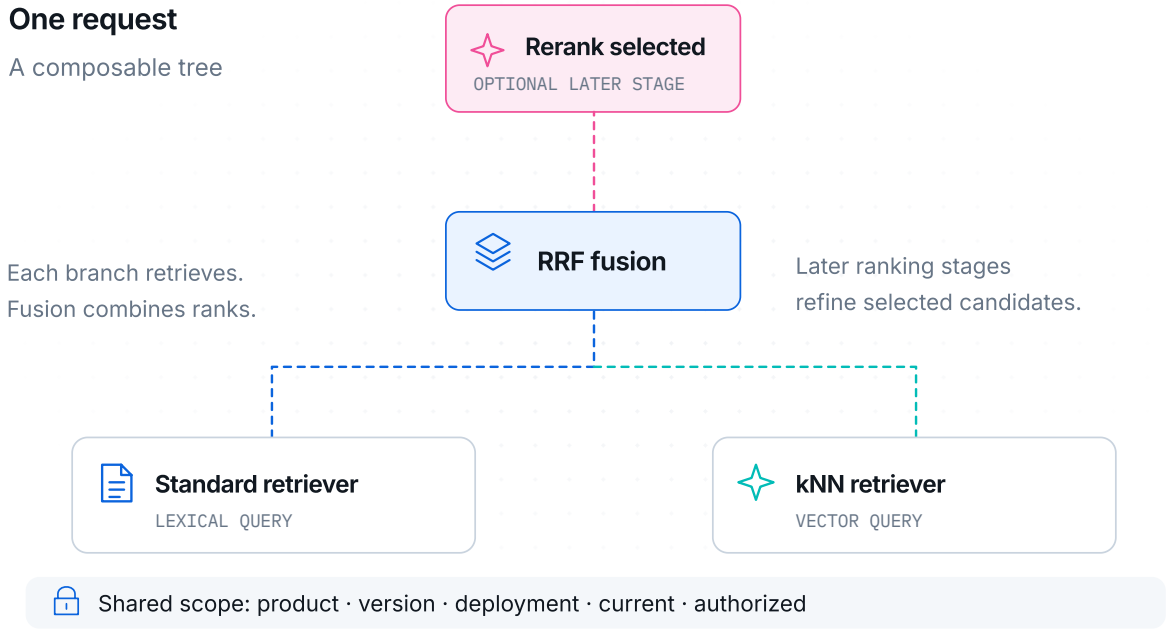
Synthetic ranks. RRF, inference, and other retrievers have version and subscription requirements. Validate retrieval windows on your workload.

SOURCES RRF ↗ Retrievers ↗

elastic

Design retrieval as a pipeline.

Retriever trees express retrieval, fusion, later ranking, and explicit rules in one search request. The application still owns the policy and evaluation.



RETRIEVE

Eligible candidates

Compose standard and vector branches with shared scope.

RANK

Successive decisions

Fuse, then selectively rerank candidate passages.

RULES

Explicit policy

Keep pinning and exclusion decisions testable.

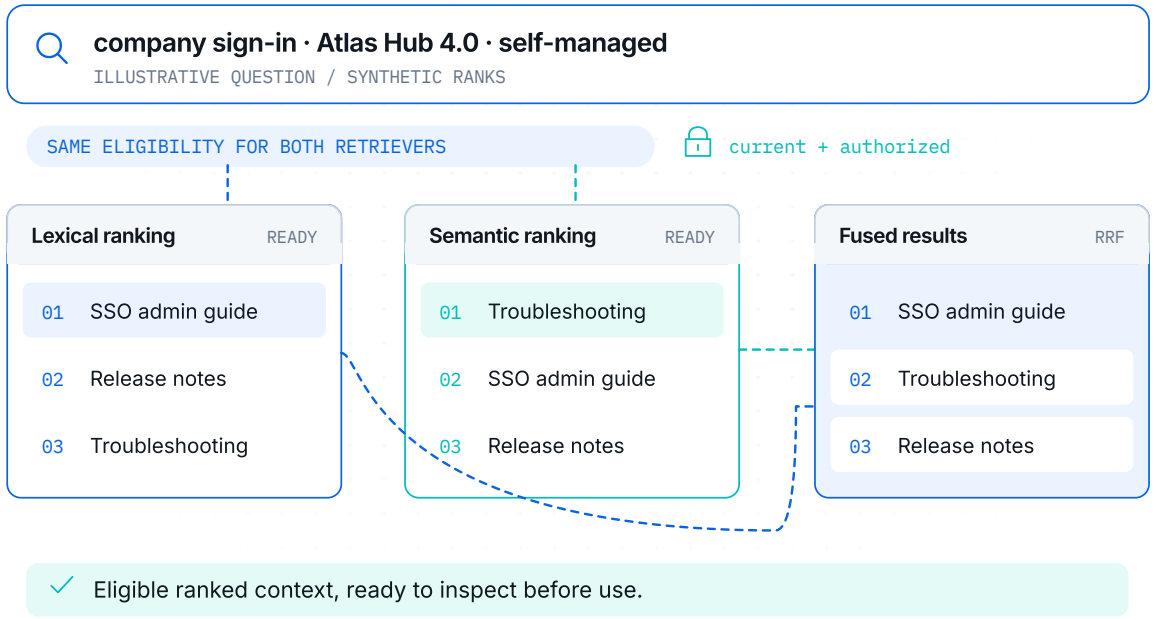
Supported retrievers, nesting, syntax, and entitlements depend on the release. Validate the actual request against your deployment.

SOURCES Retrievers ↗ Query rules ↗

elastic

Follow the question. Watch context converge.

The same scoped question travels through lexical and semantic retrieval, then returns one fused list of eligible current sources.



Fused results 4/4

QUESTION

One consistent scope

The natural-language request keeps the exact deployment.

METHODS

Different perspectives

Retrieve eligible candidates through complementary signals.

RESULT

Current sources

Administrator guide, troubleshooting, and release notes.

Illustrative UI, synthetic rankings, and no live query. Excluded inputs never become hits, snippets, context, or citations.

SOURCES RRF ↗ Retrievers ↗

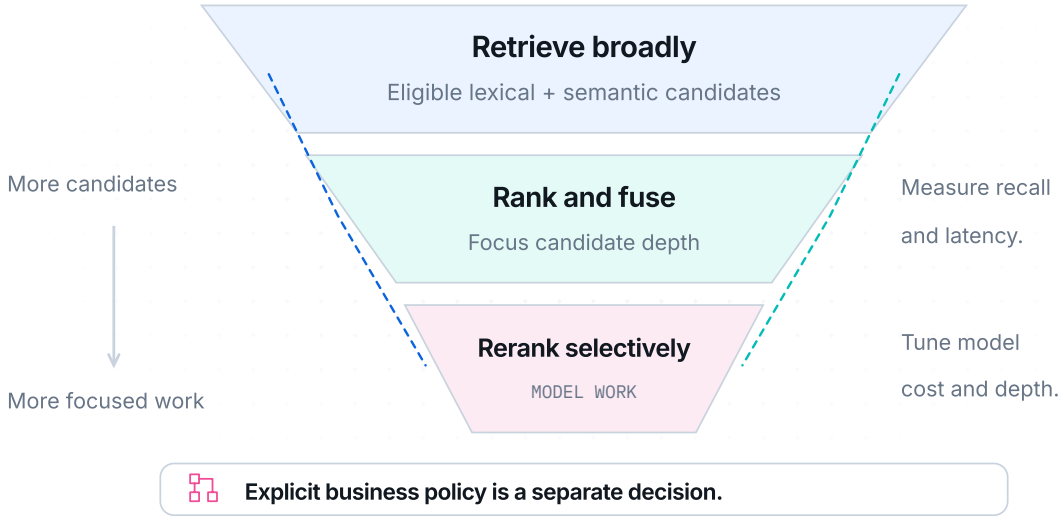
elastic

Retrieve broadly. Rerank selectively.

Use efficient retrieval to find candidates, then spend more work where it changes the result. Keep eligibility and business policy explicit.

Candidate depth and resource use

CONCEPTUAL TIERS / NO QUANTITATIVE PERFORMANCE CLAIM



Rerank selectively 3/3

COVERAGE

Find the evidence

A later stage cannot recover a document never retrieved.

PRECISION

Spend work deliberately

Limit reranking to an evaluated candidate window.

POLICY

Separate decisions

Authorization and business rules remain explicit.

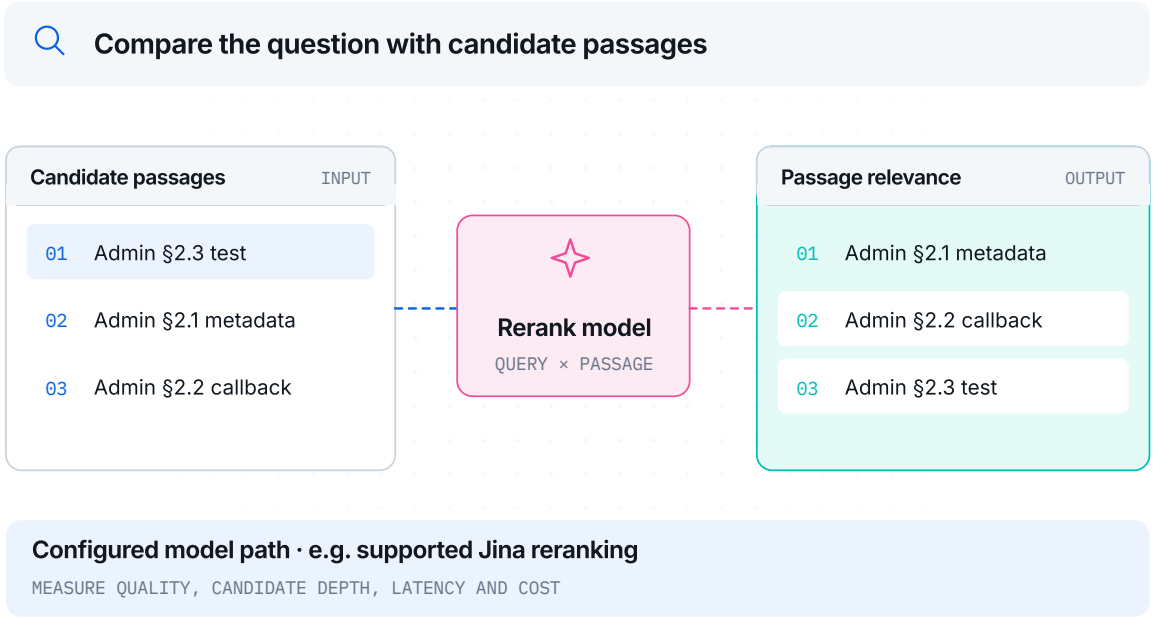
Conceptual pipeline. No universal optimum or latency claim. Later ranking can only use candidates supplied to it.

SOURCES Ranking ↗ Reranking ↗

elastic

Compare the question with the passage.

A configured rerank model assesses candidate text against the query. Make the most useful prerequisite passage easier to find.



Focused context 3/3

COMPARE

The specific task

Assess passage relevance to this question.

CONFIGURE

A supported endpoint

Choose a model and hosting path that fit the boundary.

MEASURE

Quality and work

Evaluate rerank depth, latency, throughput, and cost.

Illustrative ordering. Model, API, hosting, hardware, version, and commercial requirements apply. No guaranteed relevance improvement.

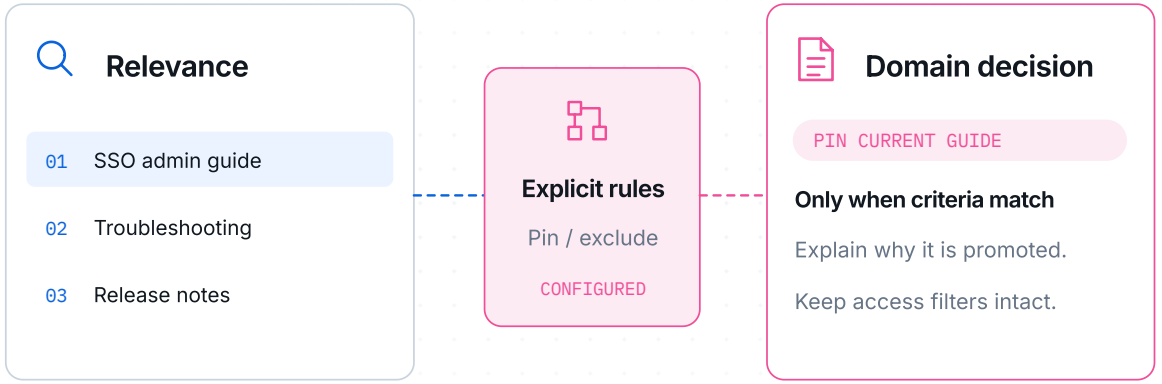
SOURCES Reranking ↗ Jina models ↗

elastic

Business rules are explicit decisions.

Pin or exclude eligible documents when configured criteria match. Keep policy separate from the relevance signals it deliberately changes.

Relevance and policy answer different questions.



Policy promotion does not make a passage semantically relevant.

CRITERIA

A stated trigger

Apply a rule to declared query metadata.

ACTION

Pin or exclude

Make the rule's result visible and reviewable.

EVALUATION

Test the consequence

Check affected queries and rule ordering.

Rules do not replace authorization. Configure privileges, criteria, ruleset order, supported retriever composition, and subscription requirements.

SOURCES Query rules ↗ Retrievers ↗

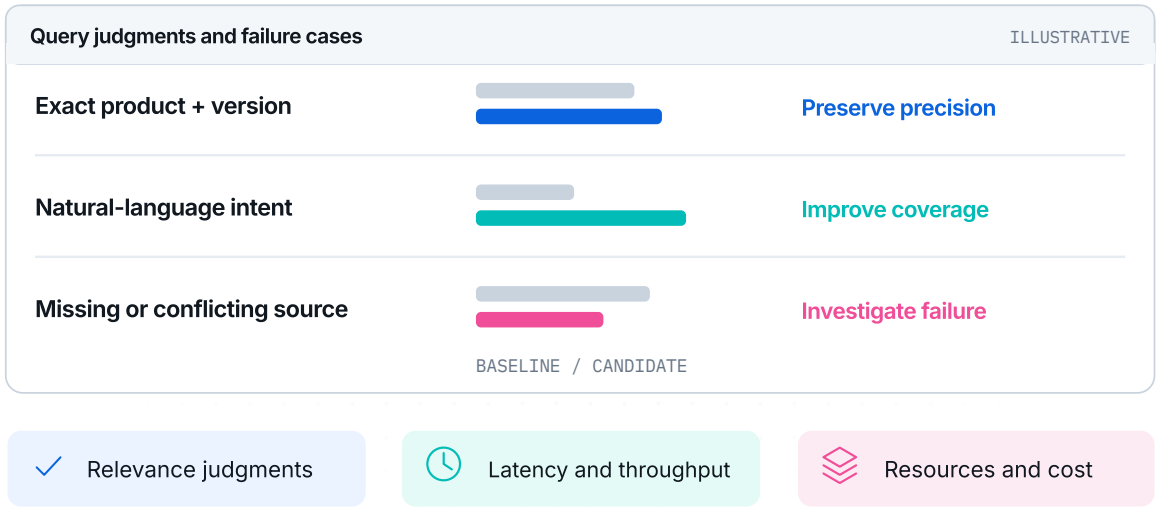
elastic

Relevance is demonstrated.

Use representative questions, rated documents, a stable baseline, and failure cases. Evaluate retrieval offline, then test the user experience.

Evaluate a representative query set

SYNTHETIC ILLUSTRATION / NO PRODUCT BENCHMARK



JUDGMENTS

Representative coverage

Rate exact, natural-language, version, and access cases.

BASELINE

A fair comparison

Hold corpus, query set, judgments, and cutoff consistent.

RELEASE

Quality with limits

Check relevance, latency, cost, and excluded-content failures.

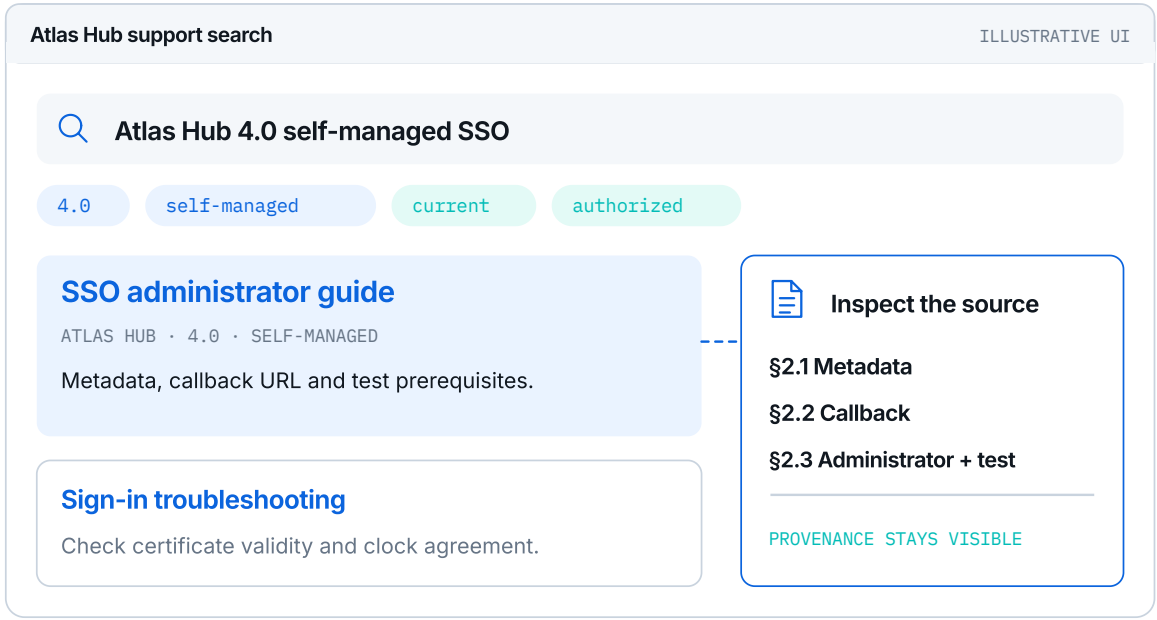
Illustrative evaluation, not measured product results. Define dataset, judgments, metric, cutoff, baseline, and method before reporting an improvement.

SOURCES Rank evaluation ↗ Judgment lists ↗

elastic

A result earns trust when it can be inspected.

Show source titles, version and deployment metadata, useful passages, and filters. Let the user verify that the result fits the task.



CONTEXT

Applicability is visible

Show current 4.0 and self-managed metadata.

EVIDENCE

Inspect the source

Preserve snippets and guide-section provenance.

FEEDBACK

Improve the experience

Use reviewed signals alongside relevance judgments.

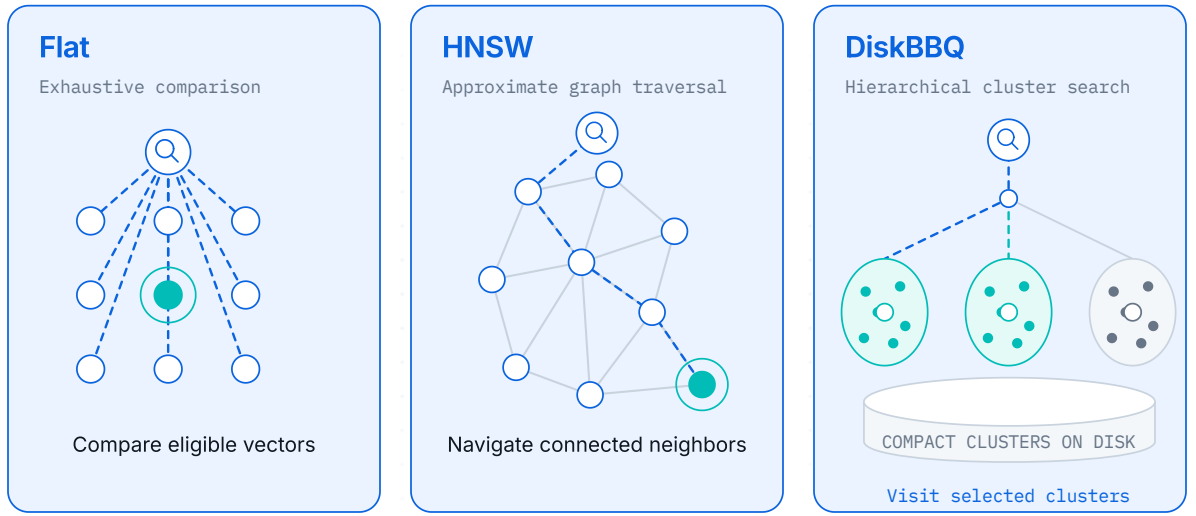
Illustrative UI, not a shipped Elastic interface. All shown hits and snippets are eligible current fictional support documents.

SOURCES Search ↗ Judgment lists ↗

elastic

Vector search starts with a workload choice.

Flat, HNSW, and DiskBBQ use different search mechanisms. Choose with your corpus, filters, recall target, memory, and query load.



Choose by recall, filters, corpus, concurrency and hardware.

DiskBBQ 3/3

FLAT

Exhaustive comparison

Useful when the eligible candidate population is small.

HNSW

Graph traversal

Evaluate approximate retrieval against the recall target.

DISKBBQ

Disk cluster search

Trade searched cluster scope against recall and work.

Qualitative illustration; no measured speed or recall claim. DiskBBQ requires Enterprise. Index behavior and defaults depend on the release.

SOURCES BBQ / DiskBBQ ↗ Capacity design ↗

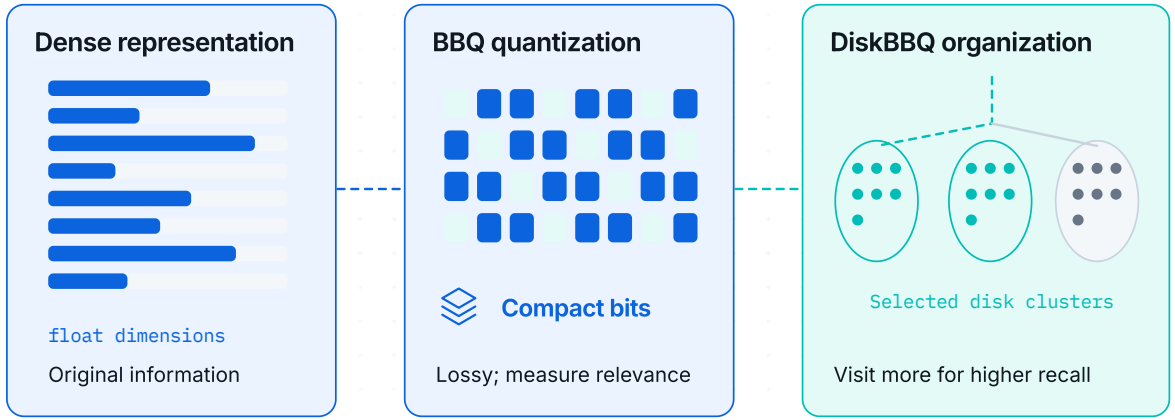
elastic

Smaller representations. Different search paths.

Quantization reduces the vector working set. Disk indexing changes how candidates are searched. They can work together, with measured trade-offs.

Two engineering decisions

A smaller representation is different from a disk-based search index



Original + quantized vectors remain on disk.

Benchmark recall, working memory, latency and total storage.

Disk cluster search 3/3

QUANTIZE

A compact representation

Lossy compression changes precision and working size.

STORE

Memory and disk boundaries

Account for original and compact representations.

VERIFY

Quality with efficiency

Evaluate recall, rescoring, latency, and total cost.

No universal savings claim. Quantization is lossy; index format, dimensions, rescoring, version, subscription, and hardware requirements apply.

SOURCES BBQ / DiskBBQ ↗ Capacity design ↗

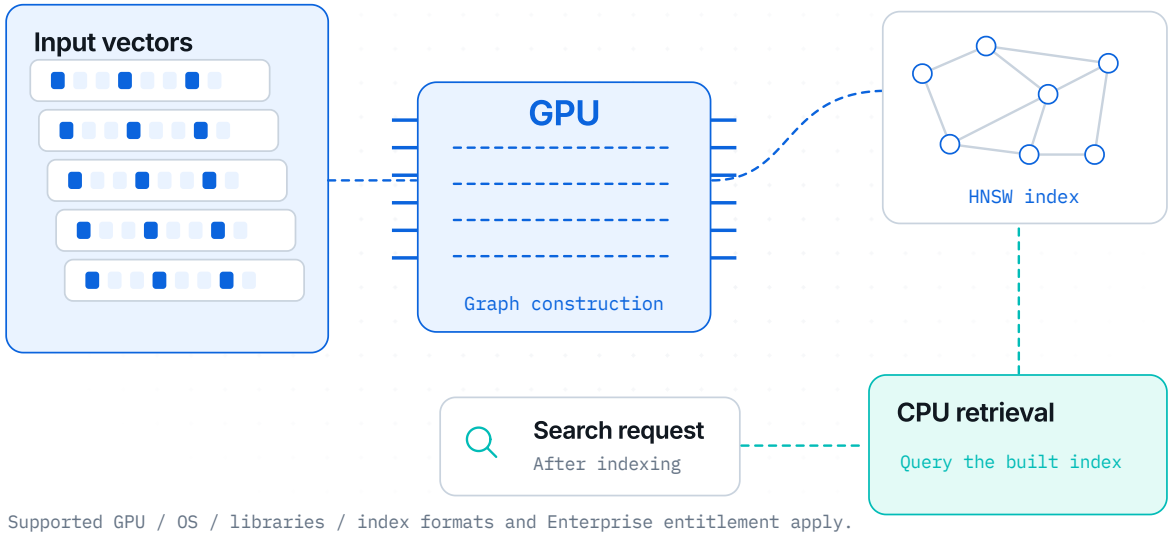
elastic

Build the graph with the GPU.

GPU vector indexing accelerates supported HNSW graph construction. Subsequent query retrieval runs on the CPU; inference is a separate workload.

Accelerate construction; keep the roles clear

Illustrative dense-vector ingest and retrieval paths



Supported GPU / OS / libraries / index formats and Enterprise entitlement apply.

CPU query retrieval 3/3

BUILD

Graph construction

Assess bulk ingest and rebuild operations.

SEARCH

CPU query path

Measure retrieval while indexing and merging.

FIT

Supported infrastructure

Validate GPU, OS, Java, libraries, and index format.

Enterprise and supported NVIDIA GPU, Linux, Java, libraries, and index format required. No universal acceleration or cost claim.

SOURCES GPU indexing ↗

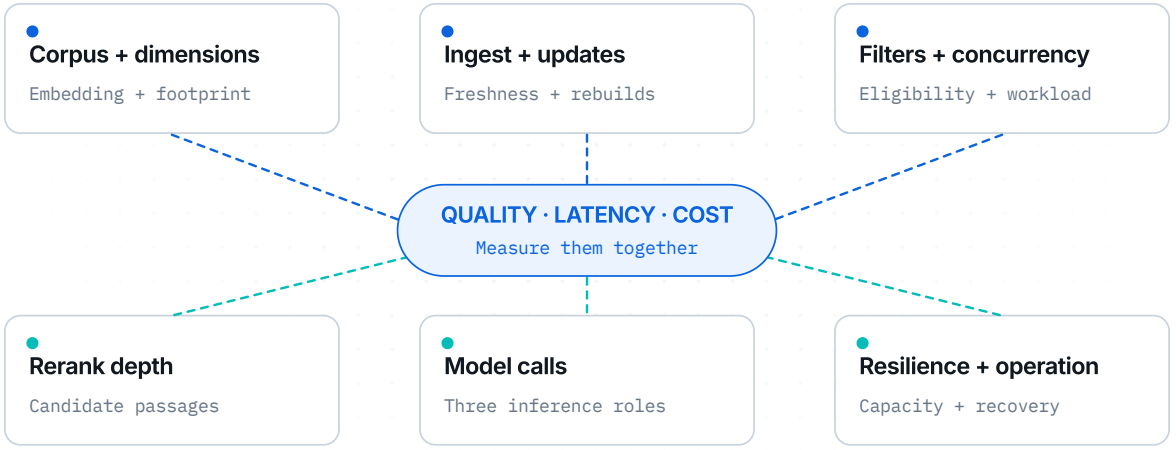
elastic

Scale the workload. Budget the whole system.

Corpus growth, updates, filters, concurrency, candidate windows, rerank depth, and model calls all change the operating envelope.

Design for the whole workload

Operating economics includes retrieval, inference, infrastructure and people



Benchmark your corpus, queries, hardware and service boundaries.

PLATFORM

Ingest and retrieval

Measure capacity, resilience, filters, and update rate.

MODELS

Inference work

Account for embeddings, reranking, and generation.

OPERATIONS

The full responsibility

Include subscriptions, hardware, people, and recovery.

Workload guidance, not a price estimate or guaranteed saving. Entitlements, service billing, hardware, and operating responsibilities vary.

SOURCES Capacity design ↗ Subscriptions ↗

elastic

Choose the model for the job.

Sparse or dense retrieval, multilingual text, compatible multimodal models, reranking, and generation solve different parts of the experience.

Select a model for its job

Representation, reranking and generation have different responsibilities

Sparse / dense

Text representation

ELSER / configured embedding models

Language/model fit

Multimodal

Text + image inputs

Dedicated field + compatible endpoint

Input / API fit

Reranking

Query × passage

Configured Jina or supported model

Depth / latency fit

Generation

Evidence → answer

Configured LLM local / approved route

Quality / trust fit

Validate mapping, model, language, hosting and hardware requirements.
A provider change does not make every embedding model interchangeable.

RETRIEVE

Represent the corpus

Choose sparse or dense models suited to language and domain.

RERANK

Compare candidates

Use a supported query-passage inference endpoint.

GENERATE

Compose from evidence

Configure a compatible model with an explicit boundary.

semantic_text is text-only. Language, modality, mapping, API, hardware, license, and deployment requirements vary; model changes may require reindexing.

SOURCES ELSER ↗ Jina models ↗ Semantic fields ↗

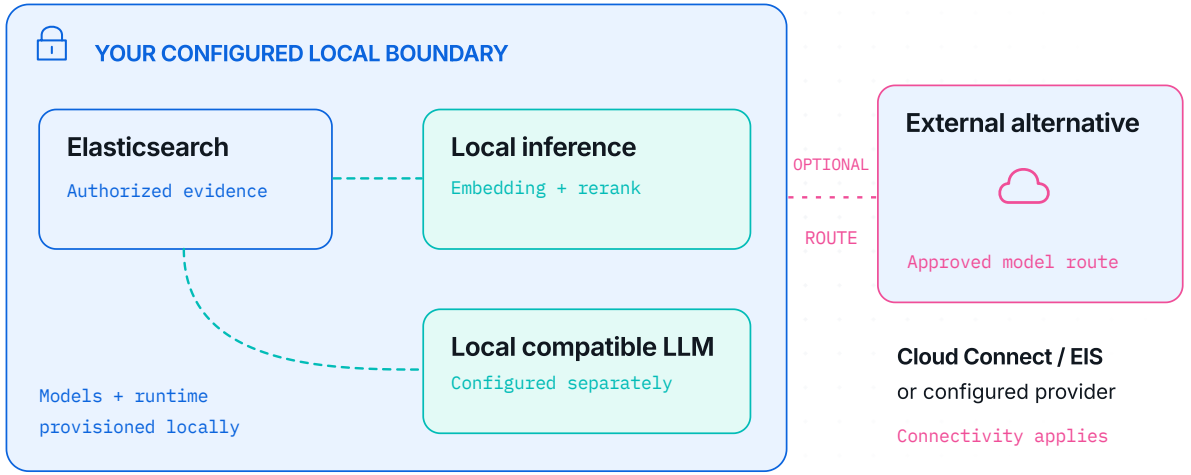
elastic

Your data boundary includes inference.

A self-managed engine can use configured local inference. An approved external model is a separate architecture choice with explicit network and data flows.

Make the inference boundary explicit

A self-managed cluster and a model endpoint are separate deployment decisions



Local model licensing and capability vary; EIS is a connected cloud service.

Approved external alternative 3/3

LOCAL

Provision the boundary

Prepare model artifacts, runtime, entitlement, and operations.

EXTERNAL

Approve the connection

Specify provider, context transfer, network, and billing.

CONTROL

Verify the actual path

A local cluster alone does not make inference local.

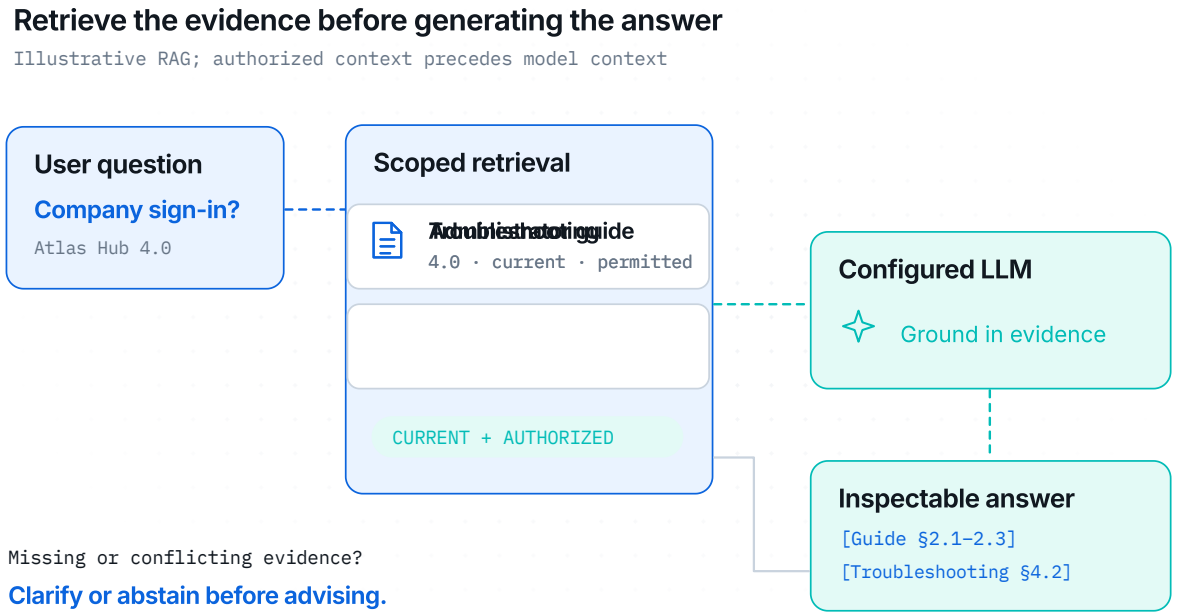
Model compatibility and commercial requirements apply. Self-managed has no preconfigured Agent Builder model. Validate restricted/offline operation before use.

SOURCES Jina On-Prem ↗ Self-managed EIS ↗ Agent models ↗

elastic

Answers begin with authorized context.

Retrieve current evidence, assemble a bounded context, generate an answer, and expose citations. Keep unsupported or conflicting claims visible.



Inspect citations 4/4

RETRIEVE

Eligible evidence first

No restricted or wrong-deployment content reaches the model.

GENERATE

Evidence-led response

Use source passages to support the answer.

VERIFY

Inspect and qualify

Clarify or abstain when evidence is missing or conflicting.

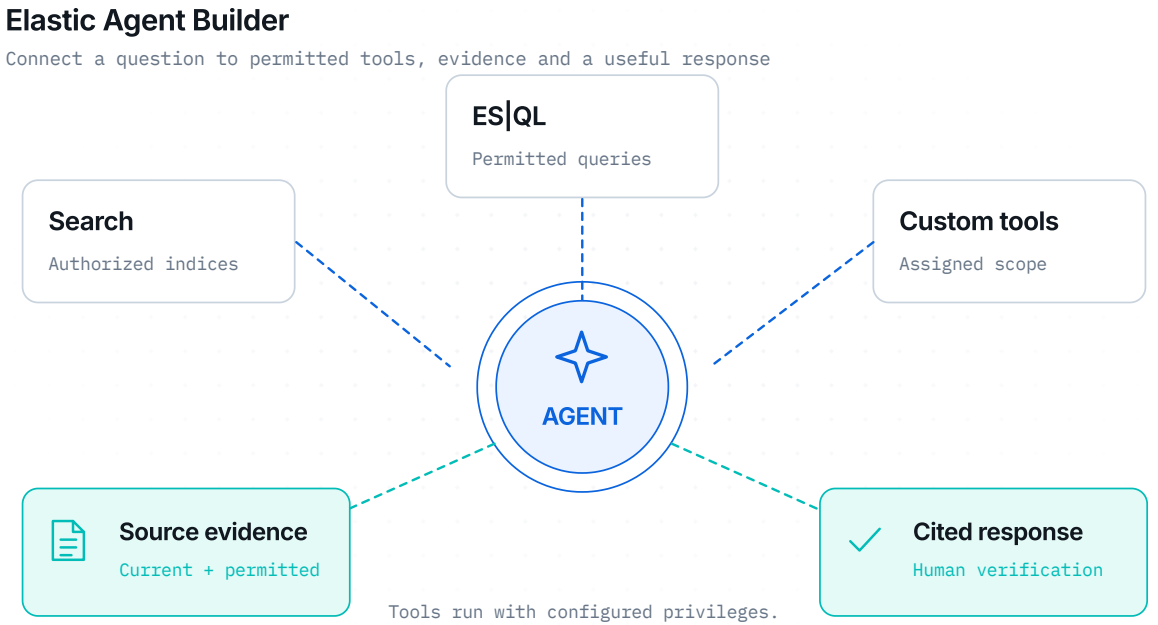
Synthetic answer and no live model call. Grounding and citations do not guarantee factual correctness or automatic authorization.

SOURCES RAG ↗ Grounding ↗ Document access ↗

elastic

Elastic Agent Builder. Evidence before action.

Connect reasoning to permitted retrieval tools over Elasticsearch data. Make scope, tool privileges, model configuration, and evidence inspection explicit.



TOOLS

A limited capability

Assign the retrieval operations needed for the task.

PRIVILEGES

The actual user scope

Enforce Kibana, Elasticsearch, and space permissions.

EVIDENCE

A reviewable response

Inspect tool results, source sections, and qualifications.

Subscription and version requirements apply. Self-managed needs a configured model. Tools execute within configured privileges; prompts are not access controls.

SOURCES Agent Builder ↗ Agent permissions ↗ Agent models ↗

elastic

One question.
A traceable answer.

Which prerequisites apply, and which guide section supports each one?

Follow the evidence, then answer

Fictional Atlas Hub journey · no tools or models are called

1. Ask the follow-up
2. Scope the tool
3. Retrieve evidence
4. Link prerequisites
5. Qualify the answer

ATLAS HUB · ILLUSTRATIVE AGENT

Prerequisites with inspectable sources

- ✓ **Identity-provider metadata** and signing certificate — Guide §2.1
- ✓ **Registered self-managed** callback URL — Guide §2.2
- ✓ **Authorized administrator** and permitted test account — Guide §2.3

If the test fails: certificate and clock checks [§4.2].

Missing deployment detail or conflicting evidence? Clarify.

Qualified answer 5/5

SCOPE

The same request

Follow-up keeps product, release, deployment, and access fixed.

SUPPORT

Claim to section

Metadata, callback, and test prerequisites have sources.

LIMIT

No hidden action

An answer does not configure or enable sign-in.

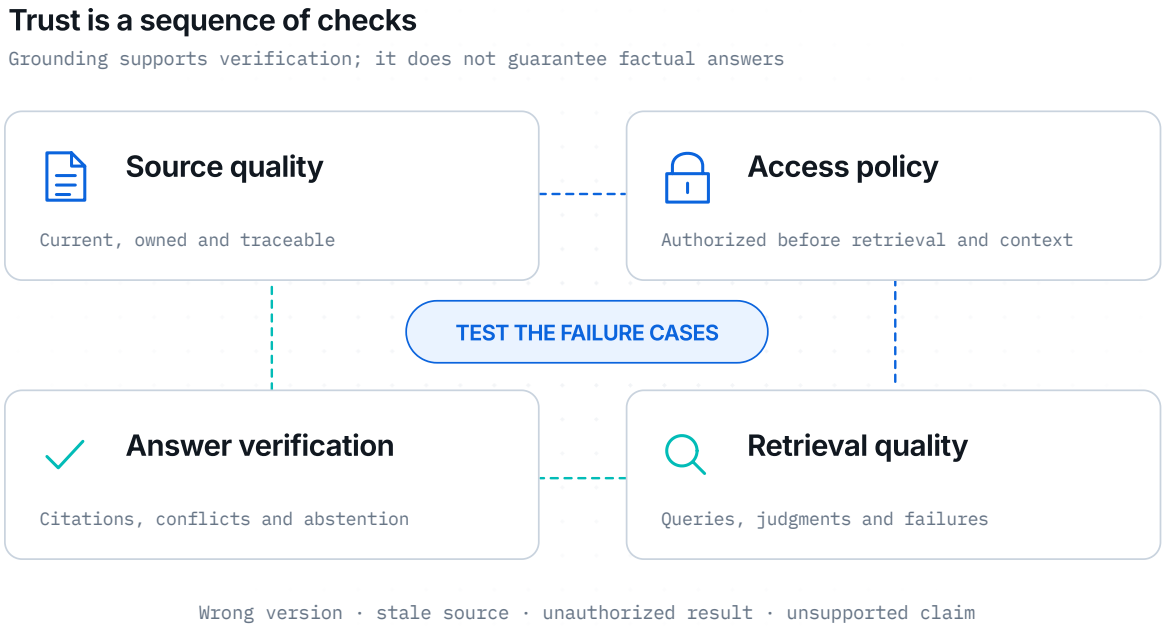
Synthetic interface, documents, answer, and tool trace. No live model or tool call. Only eligible current sources appear in context and citations.

SOURCES Agent Builder ↗ Agent permissions ↗ RAG ↗

elastic

Trust is a system you can verify.

Source quality, retrieval quality, permissions, and generated-answer behavior are separate checks. Design for uncertainty rather than hide it.



SOURCES

Current and applicable

Check revision, scope, provenance, and conflicts.

RETRIEVAL

Relevant and authorized

Test coverage, access boundaries, and source leakage.

ANSWERS

Supported and qualified

Evaluate citations, unsupported claims, and abstention.

Recommended application controls; not a guarantee of zero hallucinations or automatic complete governance. Grounding is one layer of verification.

SOURCES Grounding ↗ Agent permissions ↗ Document access ↗

elastic

Run where requirements lead.

Self-managed, ECK, ECE, Elastic Cloud Hosted, and Serverless offer different control and operating models. Compare the requirements that matter to your platform.

Choose the deployment for the requirements

Check model routes, authentication, lifecycle and feature support for each option

DEPLOYMENT	CONTROL MODEL	OPERATING RESPONSIBILITY
Self-managed	Your infrastructure	Your operations
ECK	Kubernetes operator	Your Kubernetes estate
ECE	Deployment orchestration	Your platform estate
Cloud Hosted	Managed clusters	Configured cluster choices
Serverless	Managed infrastructure	Automatic service updates

Map control, capabilities, connectivity and operations before choosing.

SELF-MANAGED

Direct responsibility

Own infrastructure, upgrades, resilience, and model routes.

ECK / ECE

Orchestrated estates

Operate Kubernetes or deployment fleets with explicit owners.

CLOUD

Managed options

Assess Hosted or Serverless capabilities and connectivity.

No blanket feature parity. Deployment, version, subscription, operations, connectivity, and model requirements differ.

SOURCES Deployments ↗ Agent models ↗ Subscriptions ↗

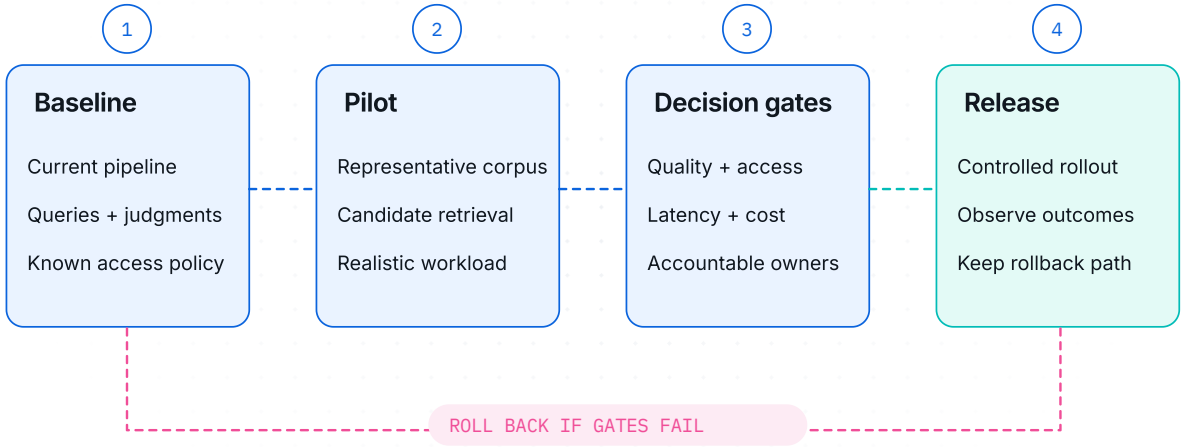
elastic

Evolve the stack. Prove each release.

Keep the existing baseline, pilot a candidate pipeline, verify quality and access gates, then roll out or return to the known path.

Evolve the stack with evidence

The next release earns its place against your current baseline



Agree thresholds and owners before the pilot starts.

Rollout or rollback 4/4

PRESERVE

A rollback baseline

Make the comparison and recovery path reproducible.

VALIDATE

Explicit release gates

Test judgments, exclusions, workload limits, and answers.

OPERATE

Named owners

Assign source, relevance, model, and platform responsibility.

Recommended rollout method. Integration and migration work are deployment-specific; no automatic parity or migration-success claim.

SOURCES Rank evaluation ↗ Deployments ↗

elastic

Evidence from customers. A benchmark for you.

HSE reports a 4% rise in search-page click-through rate. Stack Overflow's public account describes hybrid retrieval and answers linked to sources.

Customer evidence to inform an evaluation

Public customer accounts; results are examples, not promises

HSE

4%

rise in search-page click-through rate

Customer-reported A/B result

Elastic Cloud on AWS

elastic.co/customers/hse

Stack Overflow

Lexical → hybrid retrieval



Answers with attribution

The public account describes trusted community and Teams knowledge as answer context.

elastic.co/customers/stack-overflow

Different environments and workflows require their own measured evaluation.

HSE

4% rise in search-page CTR

Customer-reported result from Elastic Cloud on AWS.

STACK OVERFLOW

Retrieval to sourced answers

A qualitative public account of hybrid search and attribution.

YOUR PLATFORM

Evaluate your outcome

Use your corpus, users, deployment, judgments, and workload.

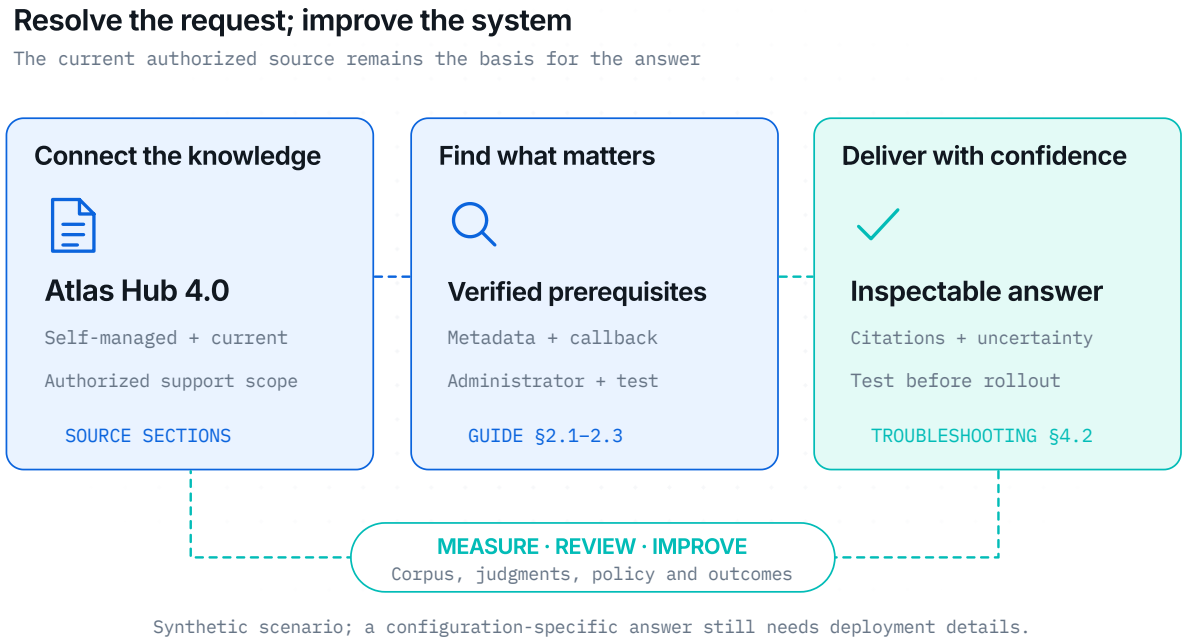
Attributed customer examples, not promises. HSE's Cloud on AWS result is not an on-premises performance or cost guarantee.

SOURCES HSE customer story ↗ Stack Overflow story ↗

elastic

The question is answered. The work keeps improving.

Maya receives current self-managed prerequisites with source sections. The team keeps the content, access policy, retrieval, and answer evaluation under review.



Synthetic scenario; a configuration-specific answer still needs deployment details.

Measured improvement 3/3

RESOLVED

Source-backed prerequisites

Metadata, callback, and test access map to current guide sections.

QUALIFIED

Known limits

A deployment-specific detail still needs clarification.

OPERATED

An accountable loop

Review changes, failures, access, and measured releases.

Fictional outcome. Review and feedback are an application operating process, not automatic persistent learning or a guaranteed improvement.

SOURCES RAG ↗ Rank evaluation ↗ Document access ↗

elastic

Connect the knowledge. Deliver with confidence.

Find what matters, prove it with your questions, and make the next platform decision from evidence. Start with a bounded evaluation.

Build the evaluation together

Agree a corpus, representative questions, judgments and access policy



Your next step: one measured Search / AI pilot.

Named owners · acceptance gates · workload and cost limits · rollback

CONNECT

Current, authorized context

Agree the corpus, source owners, and access policy.

FIND

Measured relevance

Agree representative questions, judgments, and a baseline.

DELIVER

A release you can justify

Agree workload limits, owners, gates, and rollback.

Evaluation proposal, not a commitment to savings, zero hallucinations, universal feature parity, or an unreviewed production rollout.

SOURCES Search ↗ Rank evaluation ↗

elastic